

Let $x(t) \in L^1$ admit Fourier series a_k

Transformations		Remarks
Conjugate	$\overline{x(t)} \rightarrow \overline{a_{-k}}$	if $x(t) \in \mathbb{R} \implies x(t) = \overline{x(t)} \implies a_k = \overline{a_{-k}}$
Time inversion	$x(-t) \rightarrow a_{-k}$	if $x(t)$ even $\implies x(t) = x(-t) \implies a_k = a_{-k}$
Time scale	$x(\alpha t) \rightarrow a_k \alpha > 0$	$T' = T/\alpha, w' = \alpha w$
Traslation	$x(t + \beta) \rightarrow e^{jk w_0 \beta} a_k$	
Modulation	$e^{jM w_0 t} \rightarrow a_{k-M}$	
Derivation	$\frac{dx(t)}{dt} \rightarrow jk w_0 a_k$	

Theorem 1 (Coefficients for LTI systems).

Let $x(t)$ admit coefficients a_k and let $h(t) \in L^1(\mathbb{R})$, then $y(t) = h * x(t)$ will have the following coefficients:

$$b_k = H(jw_k) a_k$$

Proof.

Simply $x(t) = \sum_{-\infty}^{\infty} a_k e^{jw_k t}$ then $H(jw)x(t) = \sum_{-\infty}^{\infty} H(jw) a_k e^{jw_k t}$

□

Theorem 2 (Coefficients of $x(t) = \text{rep}_T(\text{rect}(t/t_0))$).

Let $x(t) = \text{rep}_T(\text{rect}(t/t_0))$ with $t_0 \leq T$, then the coefficients a_k are:

$$a_k = \frac{t_0}{T} \text{sinc}\left(\frac{w_k t_0}{2\pi}\right)$$

Proof.

By definition

$$a_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-jw_k t} dt$$

Let's compute only the integral for now, and since $t_0 \leq T$

$$\int_{-\frac{t_0}{2}}^{\frac{t_0}{2}} e^{-jw_k t} dt = \frac{-1}{jw_k} (e^{-jw_k t}) \Big|_{-\frac{t_0}{2}}^{\frac{t_0}{2}} = \frac{-1}{jw_k} (e^{-jw_k \frac{t_0}{2}} - e^{jw_k \frac{t_0}{2}}) = \frac{1}{jw_k} (-e^{-jw_k \frac{t_0}{2}} + e^{jw_k \frac{t_0}{2}})$$

by using Euler's equation

$$\frac{2}{w_k} \sin(w_k \frac{t_0}{2}) = \frac{2}{w_k} \sin(\pi \frac{w_k t_0}{2\pi}) = \frac{2}{w_k} \sin(\pi \frac{w_k t_0}{2\pi}) = t_0 \text{sinc}\left(\frac{w_k t_0}{2\pi}\right)$$

where the last step was possible by multiplying by $\frac{2t_0\pi}{2t_0\pi}$

□

Theorem 3 (Parseval).

Let $x(t) = \sum a_k e^{jkw_0 t} \implies E(x(t)) = T \sum |a_k|^2$

More in particular, if $y(t) = \sum b_k e^{jkw_0 t}$ has same period as $x(t)$, then

$$E(x(t), y(t)) = T \sum a_k \overline{b_k}$$

Fourier Transform

Definition 4 (Fourier transform).

Let $x(t)$ be a signal with $t \in (R)$, then the Fourier transform is defined as

$$X(w) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Theorem 5 (Existence of $X(w)$).

$$\text{Let } x(t) \in L^1((R)) \implies \exists X(w)$$

But clearly if $x(t)$ is periodic, then $x(t) \notin L^1(R)$. For $x(t) = e^{jw_0 t}$ the fourier transform is:
 $X(w) = 2\pi\delta(w - w_0)$. Since by fourier series every periodic signal can be written as
 $x(t) = \sum_{-\infty}^{\infty} a_k e^{jw_k t}$ with $w_k = kw_0$, the clearly $X(w) = 2\pi \sum_{-\infty}^{\infty} a_k \delta(w - w_k)$